

Augmenting Telephony Audio Data using Robust Principal Component Analysis

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Abstract—Audio augmentation (e.g., corrupting audio data by noise) has been shown to improve the performance of Automatic Speech Recognition (ASR) systems for low-resource languages. In light of this, we are interested in understanding whether corrupting speech data with telephone channel characteristics (e.g., background music, artifact caused by down-sampling) improves the performance of ASR systems as well. In this work, we investigate the possibility of applying Sound Source Separation (SSS) approaches to capture the telephone channel characteristics. We are in particular interested in Robust Principal Component Analysis (RPCA), which is an unsupervised approach used for various SSS tasks. Our results show that augmenting clean speech data corpus with telephone channel characteristics yields a more robust ASR system, with 7.8% of Word Error Rate reduction. We also find that the characteristic, which has the lowest spectral features, improves ASR the most.

Index Terms—speech recognition, data augmentation, sound source separation, robust principal component analysis

I. INTRODUCTION

PREVIOUS studies have shown that data augmentation can improve the performance of Automatic Speech Recognition (ASR) systems [1]–[4]. In other words, by augmenting the existing training set with the expanded training set resulted from data augmentation, the robustness of ASR systems is improved. Traditional audio data augmentation techniques include pitch shifting, time-stretching, and playback speed modification, to name but a few. In addition to these approaches, various data augmentation approaches have been investigated. Gales *et al.* [5] showed that corrupting audio data by noise improved the robustness of ASR. They evaluated their idea using discriminative classifiers such as Support Vector Machines. Ragni *et al.* [1] examined two data augmentation schemes on two low-resource languages. They found that data augmentation yielded a consistent improvement of speech recognition performance by using these augmentation schemes. Ko *et al.* [2] investigated three audio-level speech augmentation methods, namely vocal tract length perturbation, tempo perturbation, and speed perturbation. Their results showed that speed perturbation based data augmentation gave

the best result. In another study conducted by the same group [3], data augmentation was applied to reverberant speech signals. They investigated the impact of adopting simulated room impulse responses (RIRs) and the effect of adding point-source noises. They found that when point-source noises were added, using simulated RIRs yields better performance, thus narrowed the performance gap between using simulated and real RIRs. In a recent study conducted by Fukuda *et al.* [4], data augmentation was adopted to improve foreign-accented speech recognition. Three augmentation approaches were studied, and both supervised and unsupervised approaches to using the accented data were examined. They found that all augmentation approaches provided improvements to some extent, and speed modification outperformed the others. More recently, Park *et al.* [6] proposed a new data augmentation approach for audio data by treating the augmentation problem as a visual problem rather than an audio one. Instead of manipulating the audio waveform, this approach amends the spectrogram to increase the number of available data. This method is rather simple, yet it showcases a significant improvement on 960-hour LibriSpeech and 300-hour Switchboard benchmarks. To summarize, the robustness, as well as the performance, of ASR systems can be improved by data augmentation.

Even though data augmentation has been investigated in various studies, and corrupting audio data by noise has been shown to improve the robustness of ASR, to the best of our knowledge, corrupting audio data by telephone channel characteristics has not yet been examined. The reason why it is important is twofold. First, telephony data can be hard to acquire. Most of them contain sensitive information. Second, annotating audio data can be time-consuming, which is a common problem in low-resource scenarios. We will save a lot of effort if there is a way to synthesize telephone-quality audio data from clean speech data that are already annotated.

In this paper, we propose an audio data augmentation method based on the Sound Source Separation (SSS) approach. That is, we separate the telephone channel characteristics (artifacts caused by downsampling, background noise, music, etc.) from telephone call recordings, and generate artificial telephony speech data by combining clean speech recordings and telephone channel characteristics at various Signal-to-Noise Ratios (SNR). We are in particular interested in unsupervised SSS approaches as audio recordings are generally

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the only data one can acquire in a low-resource scenario. In other words, no annotation can be found, or annotating is very tedious. In particular, we investigate Robust Principal Component Analysis (RPCA), an unsupervised SSS approach which has been shown to solve several SSS problems [7], [8]. To evaluate our audio augmentation method, we train several ASR models on our augmented training set and see a significant improvement. We also evaluate the spectral features of the telephone channel characteristics extracted from genuine telephony speech data, trying to understand why some characteristics are more suitable for increasing the robustness of ASR.

The rest of this paper is organized as follows. Section II gives a brief introduction to RPCA and its applications for solving SSS problems. Section III provides the methodology used in this paper. Section IV gives the details of our experiment and discussion on the results, especially those on spectral features. Finally, Section V concludes our work and discusses some possible future work.

II. ROBUST PRINCIPAL COMPONENT ANALYSIS

RPCA was proposed by Candès et al. [9]. It extends the classical Principal Component Analysis (PCA), which is not robust against outliers and noise. Given a data matrix $M \in \mathbb{R}^{n_1 \times n_2}$ where n_1 and n_2 are the dimensions of M , the main goal of RPCA is to recover a low-rank matrix $L \in \mathbb{R}^{n_1 \times n_2}$ and a sparse matrix $S \in \mathbb{R}^{n_1 \times n_2}$ from M . By adopting the Principal Component Pursuit (PCP), it can recover L and S by solving

$$\begin{aligned} & \text{minimize} \quad \|L\|_* + \lambda \|S\|_1 \\ & \text{subject to} \quad L + S = M, \end{aligned}$$

where $\|L\|_*$ denotes the nuclear form of L , $\|S\|_1$ denotes the l_1 -norm of S , and λ is a weight parameter for adjusting the balance between the rank of L and the sparsity of S . For the value of λ , $1/\max(n_1, n_2)$ is a good starting point, and it can be fine-tuned to achieve better results for different applications [9]. Also, since RPCA is an unsupervised model that requires no prior information, it is very suitable for tackling SSS problems in low-resource scenarios. Note that RPCA here can be regarded as a regularization approach.

RPCA has been used in various applications for SSS problems. Huang et al. [7] used RPCA to separate singing-voice from the corresponding music accompaniment in monaural recordings. In the context of an audio signal, a spectrogram (i.e., a time-frequency representation of the original signal) can be considered as a non-negative matrix. By applying RPCA to the spectrogram, a low-rank matrix and a sparse matrix is expected to be recovered. When it comes to a musical context, a singing-voice can be considered as having more variation but relatively sparse, whereas the music accompaniment can be considered of more low-rank because of the repeating nature of music accompaniment. Huang et al. have examined different values of λ , as well as the effect of binary masking, for singing-voice separation. Their results showed that RPCA achieved around 1 ~ 1.4 dB higher Global Normalized Source

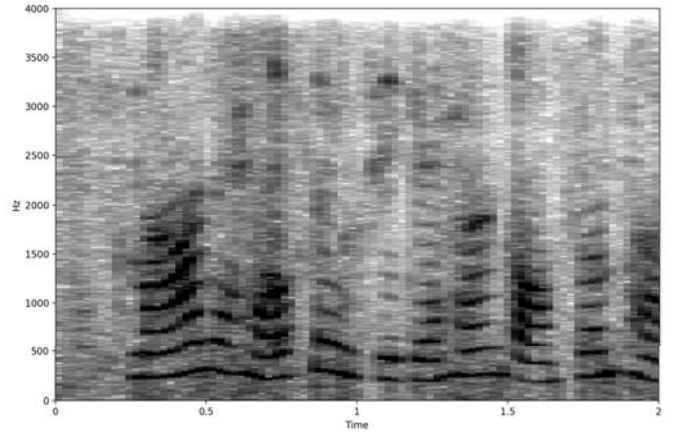


Fig. 1. Spectrogram of a telephony speech data.

to Distortion Ratio (GNSDR) compared with two state-of-the-art methods by that time. Huang et al. [8] proposed an unsupervised framework for speech denoising based on RPCA. Their results showed that their proposed method outperformed the original RPCA and Non-negative Factorization (NMF) by a higher Signal-to-Distortion Ratio (SDR) of ~ 2.48 dB and 3.23 dB since the proposed method extracted the speech part which was less polluted by noise.

In this work, we examine whether augmenting telephony speech data can improve the performance of ASR as discussed in Section I. Fig. 1 gives an example spectrogram of a telephone recording. It clearly shows the distinction between speech signal (i.e., sparse), which are the dark areas at the bottom, and the telephone channel characteristics (i.e., low-rank), which are located in the upper part. By applying RPCA to telephone recordings, we expect to capture the telephone line characteristics in those recordings.

III. MATERIALS AND METHODS

A. Telephony Speech Data Generation

To capture the telephone channel characteristics, genuine telephone recordings in Cantonese were obtained. We apply RPCA to these spectrograms to yield a low-rank matrix L , as well as a sparse matrix S . We then generate ideal binary masks M_b using both L and S :

$$M_b(m, n) = \begin{cases} 1 & |L(m, n)| > |S(m, n)| \\ 0 & \text{otherwise,} \end{cases}$$

where m ($m = 1 \dots n_1$, the total number of Fast Fourier Transform (FFT) bins used in STFT in this case) is the FFT index, and n ($n = 1 \dots n_2$, the total number of frames in this case) is the frame index. Finally, this binary mask is element-wise multiplied to the original spectrogram and transformed back to a time-domain audio signal by inverse STFT using the original phase information.

We try different λ values and $\lambda = 0.5/\max(n_1, n_2)$ gives the best balance between the low-rank matrix and the sparse

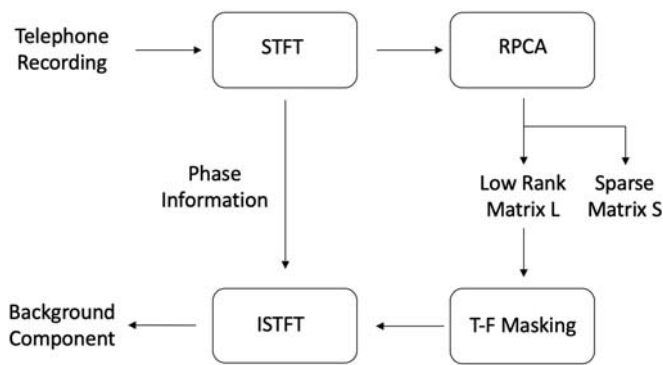


Fig. 2. Proposed framework for capturing the telephone line characteristics in telephone recordings.

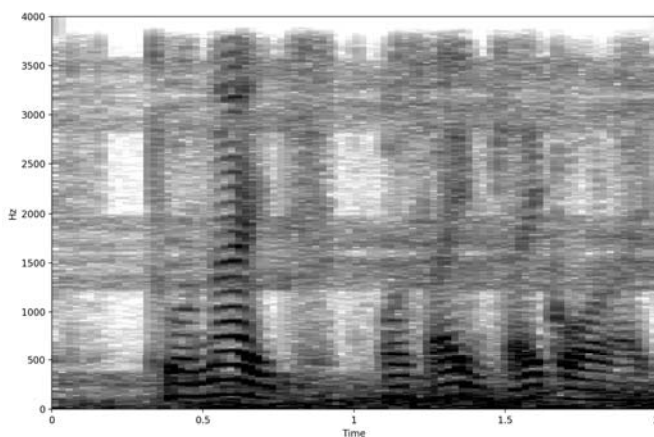


Fig. 3. Spectrogram of an artificial telephony speech data.

matrix [7]. To solve the problem effectively, the inexact Augmented Lagrange Multiplier (ALM) method is used as suggested by [7], [10]. We implement RPCA using Python. Fig. 2 depicts our approach.

To augment telephony speech data, we first create a corpus of clean speech recordings. We generate several artificial telephony speech datasets for evaluation. Each clean signal is mixed with each characteristic we computed in the previous step at 0dB SNR. Fig. 3 shows the spectrogram of an artificial telephony speech data example. Note that the vocal parts (i.e., the dark areas) are preserved, while the telephone channel characteristics are also represented in the upper part.

B. Speech Recognition System

We evaluated our dataset using an end-to-end ASR system [11] developed based on bi-directional Long Short-Term Memory (bLSTM) [12] with Connectionist Temporal Classification (CTC) [13] incorporated. Our ASR system uses filterbank and pitch features as inputs. For filterbank features, 80 coefficients are used.

TABLE I
DETAILS OF CHANNEL CHARACTERISTICS.

Characteristic	WER
1	Male voice
2	Female voice
3	Silence
4	Background music
5	Street noise

TABLE II
EVALUATION RESULTS FOR THE ASR MODELS.

Model	Training Set	WER
1	Baseline (King ASR + Clean)	36.9
2	King ASR + Characteristic 1	35.2
3	King ASR + Characteristic 2	35.3
4	King ASR + Characteristic 3	34.9
5	King ASR + Characteristic 4	35.2
6	King ASR + Characteristic 5	35.2
7	King ASR + Street Noise	35.2

IV. RESULTS

A. Experiment overview

For training and testing our ASR system, the King ASR Hong Kong Cantonese Speech Recognition Corpus is used [14]. It contains 96.7 hours of Cantonese telephony speech recordings. 51 different native Cantonese speakers, including 25 males and 26 females, were recruited. These telephony speech recordings are manually annotated by native Cantonese transcribers.

For the evaluation, a total of 87 hours from the King ASR corpus are used as training data, while the remaining 9.7 hours are used as the test set. Data in the test set are not seen and used in the training phase. For the baseline model, the training data derived above are combined with a clean speech recording corpus we prepare to form a training dataset. The recordings in this clean speech corpus are recorded by 1445 individual speakers who are native Cantonese speakers. About half of them are males and the remaining half are females. The average age is 25.4. Speakers are asked to record 215 to 300 utterances containing mainly Cantonese and mixed with some simple English phrases. Each clean speech recording is recorded using a smart phone and contains only one speaker. These audio recordings are all manually annotated by native transcribers. The average duration for the recordings is 5.2 seconds. All speech recordings are downsampled to 8kHz. In total, this dataset contains about 486 hours of clean speech recordings. Therefore, a total of 573 hours are used for training the baseline model.

B. Experiment 1

To evaluate our data augmentation approach, we first obtain five genuine telephone recordings in Cantonese, with duration ranging from 3 to 13 seconds. Only one speaker is present in each recording. These recordings are recorded from both wired and wireless telephone systems at 8kHz sampling rate.

TABLE III
EVALUATION RESULTS FOR THE ASR MODELS WITH VARIOUS SNRS.

Model	WER
King ASR + Characteristic 3 (-5dB)	35.3
King ASR + Characteristic 3 (5dB)	35.0
King ASR + Characteristic 3 (0dB)	34.9

TABLE IV
EVALUATION RESULTS FOR THE ASR MODELS WITH VARIOUS TRAINING SET.

Model	WER
King ASR + Clean + Characteristic 3	34.0
King ASR + Clean + Street Noise	34.5

The spectrogram of these signals is then computed by Short-Time Fourier Transform (STFT) with a 1024 window size and a 256 hop size. We follow the idea illustrated in Section III-A to generate five telephone characteristics, which cover different scenarios. Table I shows the details of these characteristics. These five characteristics are used to mix with each recording in the clean speech corpus, ending up with five artificial telephony speech recordings created. These artificial recordings are used for creating five new speech data corpora. We then combine each of these corpora with the King ASR dataset to yield five different training sets. We train five different models on each of these training sets.

To compare with the existing audio augmentation approaches, we also prepare a training set comprising the King ASR training data and the clean speech recording corpus corrupted by genuine street noise acquired from [15]. We train all models for 15 epochs. Table II shows the evaluation results. As we can see from the table, Model 4, which incorporates Characteristic 3, yields the best Word Error Rate (WER). WER is a metric widely used to compare the accuracy of the transcripts produced by an ASR system. WER is defined as:

$$WER = \frac{S + I + D}{N}, \quad (1)$$

where S is the number of substitutions, I is the number of insertions, D is the number of deletions, and N is the number of words in the reference (i.e., the transcript). It produces a 5.4% relative WER reduction comparing with the baseline model. Note that Model 4 even outperforms Model 7, which makes use of the genuine street noise to corrupt clean speech recordings. We will address this in Section IV-E.

C. Experiment 2

As Model 4 performs the best, we further investigate the impact on the robustness of the ASR system using Characteristic 3 only. First, we evaluate the effect of different SNR on audio augmentation. We train two additional models following the same method as mentioned above. By following [7], we create

two new training datasets with -5dB SNR and +5dB SNR, respectively. In other words, each clean signal is mixed with Characteristic 3 at both -5dB and 5dB SNR. We train two ASR models on these two training sets and use the same test set as before. Table III shows the evaluation results. Surprisingly, the performance is degraded. At -5dB and +5dB SNR, we obtain 1.1% and 0.29% relative WER boosts. We believe that this is due to the λ value we picked as it gives a good balance between the low-rank matrix and the sparse matrix.

D. Experiment 3

Finally, we expand our idea by combining the King ASR corpus used for the baseline model with the dataset generated from Characteristic 3 at 0dB SNR, as well as the clean speech dataset to form a new training set. We also come up with another training set which comprises King ASR corpus, the clean speech dataset mentioned above, and a copy of the clean speech dataset corrupted by street noise used above for comparison. Once again, two ASR models are trained on these two training sets, and we used the same test set as before. Surprisingly, our model yields a WER of 34.0%, whereas the model incorporated with street noise corrupted audio data only yields a WER of 34.5%. According to our results, the telephone line characteristic captured from authentic telephone recording can improve the robustness of our ASR better than realistic background noise. To sum up, our audio augmentation approach yields a 7.8% relative WER reduction. Table IV shows the evaluation results.

E. Spectral Features

Even though the telephone characteristics are extracted from genuine telephone recordings using RPCA, it is important to understand what makes Characteristic 3 outperform other characteristics. To address this question, we investigate the spectral features of these Characteristics. In general, spectral features describe the timbre, or “sound color”, of a sound. Spectral features are widely used in feature extraction for Audio Classification and Audio Analysis [16]–[18], as well as Music Information Retrieval [19], [20].

In particular, we are interested in three simple and commonly used spectral features, namely Spectral Centroid (SC), Spectral Spread (SS), and Spectral Flux (SF) [18], [21]. The definitions of these spectral features are shown as follows:

- SC is the center of gravity of the spectrum and is closely related to the perceptual brightness of a sound. The SC of the i th frame, SC_i is defined as:

$$SC_i = \frac{\sum_{k=1}^{W_L/2} kX_i(k)}{\sum_{k=1}^{W_L/2} X_i(k)} \quad (2)$$

where W_L is the number of DFT coefficients, and $X_i(k)$ is the magnitude of k th DFT coefficient of the i th audio frame. In general, the higher the SC a sound yields, the brighter a sound is perceived by human listeners. Note that SC is independent of the actual amplitude scale of a signal.

TABLE V
SPECTRAL FEATURES OF THE CHARACTERISTICS.

Characteristic	Average SF	Average SS	Average SC
1	0.244	2.348	0.446
2	0.248	2.576	0.512
3	0.221	1.609	0.263
4	0.233	1.947	0.365
5	0.222	1.829	0.317

- SS is the second central moment of the spectrum. The deviation of the spectrum from the SC has to be calculated first to calculate SS of the i th frame:

$$SS_i = \sqrt{\frac{\sum_{k=1}^{W_{L/2}} (k - SC_i)^2 X_i(k)}{\sum_{k=1}^{W_{L/2}} X_i(k)}} \quad (3)$$

It can be regarded as a measure of the bandwidth of the spectrum. Note that these two spectral features (i.e., SC and SS) are often normalized by $F_s/2$ in practice, where F_s is the sampling rate of an audio recording. After normalization, the features become in the range of 0 and 1 inclusively, which can be very useful when these features are combined with other features in audio analysis tasks.

- SF measures the spectral change between two consecutive frames:

$$SF_{(i,i-1)} = \sum_{k=1}^{W_{L/2}} (EN_i(k) - EN_{i-1}(k))^2 \quad (4)$$

where $EN_i = \frac{X_i(k)}{\sum_{l=1}^{W_{L/2}} X_i(l)}$ is the k th normalized DFT coefficient at the i th frame. The higher the SF is, the more fluctuant a sound is.

To quantify these spectral features, the average of each spectral feature over all frames is calculated. We use Python open-source package pyAudioAnalysis [22] to calculate the spectral features. Table V shows the results and Characteristic 3 yields the lowest values for all 3 spectral features. In other words, our results show that for our audio augmentation approach, the characteristic which is the least bright, more steady, and having a narrow bandwidth improves the robustness of our ASR model the most.

V. CONCLUSION

Audio augmentation can improve the performance of ASR systems for low-resource languages. In this paper, we present a novel approach to data augmentation for generating artificial telephony speech data based on SSS approaches. We show that the performance of ASR is improved by augmenting the training set using the telephone line characteristic captured from telephone recordings, with a 7.8% WER reduction. We also find that the characteristics which yield the best results have the lowest spectral features.

Of course, some potential questions are still unaddressed in this paper. Therefore, we would like to suggest a few possible future works. One possible work is to investigate how our proposed method performs in other learning models such as audio classification and speaker verification, to name but a few. Another possible work is to investigate the impact of using a combined data set on training an ASR. That is, to combine all the artificial telephony speech data generated in our study and train a model. We expect to see a positive impact on the performance of an ASR by this approach. As in this work, we find that the telephone characteristic having the least brightness, the most steadiness, and the narrowest bandwidth improves the robustness of ASR the most. However, the reason why such a characteristic will help is still not answered. Perhaps the answer to this question will be crucial for improving the performance of ASR systems, which will also be an interesting research direction.

VI. ACKNOWLEDGMENT

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REFERENCES

- [1] Anton Ragni, Kate M Knill, Shakti P Rath, and Mark JF Gales, "Data augmentation for low resource languages," in *Fifteenth Annual Conference of the International Speech Communication Association*, 2014.
- [2] Tom Ko, Vijayaditya Peddinti, Daniel Povey, and Sanjeev Khudanpur, "Audio augmentation for speech recognition," in *Sixteenth Annual Conference of the International Speech Communication Association*, 2015.
- [3] Tom Ko, Vijayaditya Peddinti, Daniel Povey, Michael L Seltzer, and Sanjeev Khudanpur, "A study on data augmentation of reverberant speech for robust speech recognition," in *Acoustics, Speech and Signal Processing (ICASSP), 2017 IEEE International Conference on*, 2017, pp. 5220–5224.
- [4] Takashi Fukuda, Raul Fernandez, Andrew Rosenberg, Samuel Thomas, Bhuvana Ramabhadran, Alexander Sorin, and Gakuto Kurata, "Data augmentation improves recognition of foreign accented speech," *Proc. Interspeech 2018*, pp. 2409–2413, 2018.
- [5] Mark JF Gales, Anton Ragni, H AlDamarki, and C Gautier, "Support vector machines for noise robust asr," in *2009 IEEE Workshop on Automatic Speech Recognition & Understanding*, IEEE, 2009, pp. 205–210.
- [6] Daniel S. Park, William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin Dogus Cubuk, and Quoc V. Le, "SpecAugment: A simple augmentation method for automatic speech recognition," in *INTER-SPEECH*, 2019.
- [7] Po-Sen Huang, Scott Deeann Chen, Paris Smaragdis, and Mark Hasegawa-Johnson, "Singing-voice separation from monaural recordings using robust principal component analysis," in *Acoustics, Speech and Signal Processing (ICASSP), 2012 IEEE International Conference on*, IEEE, 2012, pp. 57–60.
- [8] Jianjun Huang, Xiongwei Zhang, Yafei Zhang, Xia Zou, and Li Zeng, "Speech denoising via low-rank and sparse matrix decomposition," *ETRI Journal*, vol. 36, no. 1, pp. 167–170, 2014.
- [9] Emmanuel J Candès, Xiaodong Li, Yi Ma, and John Wright, "Robust principal component analysis?," *Journal of the ACM (JACM)*, vol. 58, no. 3, pp. 11, 2011.
- [10] Zhouchen Lin, Risheng Liu, and Zhixun Su, "Linearized alternating direction method with adaptive penalty for low-rank representation," in *Advances in neural information processing systems*, 2011, pp. 612–620.
- [11] L. Samarakoon, B. Mak, and A. Y. S. Lam, "Domain adaptation of end-to-end speech recognition in low-resource settings," in *2018 IEEE Spoken Language Technology Workshop (SLT)*, Dec 2018, pp. 382–388.

- [12] Alex Graves and Jürgen Schmidhuber, "Framewise phoneme classification with bidirectional lstm and other neural network architectures," *Neural Networks*, vol. 18, no. 5-6, pp. 602–610, 2005.
- [13] Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber, "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *Proceedings of the 23rd international conference on Machine learning*. ACM, 2006, pp. 369–376.
- [14] King ASR Hong Kong Cantonese Speech Recognition Corpus, "<http://kingline.speechocean.com/>," .
- [15] Justin Salamon, Christopher Jacoby, and Juan Pablo Bello, "A dataset and taxonomy for urban sound research," in *Proceedings of the 22nd ACM international conference on Multimedia*. ACM, 2014, pp. 1041–1044.
- [16] George Tzanetakis and Perry Cook, "Marsyas: A framework for audio analysis," *Organised sound*, vol. 4, no. 3, pp. 169–175, 2000.
- [17] Martin McKinney and Jeroen Breebaart, "Features for audio and music classification," 2003.
- [18] Theodoros Giannakopoulos and Aggelos Pikrakis, *Introduction to Audio Analysis: a MATLAB® Approach*, Academic Press, 2014.
- [19] Chang-Hsing Lee, Jau-Ling Shih, Kun-Ming Yu, and Hwai-San Lin, "Automatic music genre classification based on modulation spectral analysis of spectral and cepstral features," *IEEE Transactions on Multimedia*, vol. 11, no. 4, pp. 670–682, 2009.
- [20] Juan Pablo Bello, Laurent Daudet, Samer Abdallah, Chris Duxbury, Mike Davies, and Mark B Sandler, "A tutorial on onset detection in music signals," *IEEE Transactions on speech and audio processing*, vol. 13, no. 5, pp. 1035–1047, 2005.
- [21] James W Beauchamp, *Analysis, synthesis, and perception of musical sounds*, Springer, 2007.
- [22] Theodoros Giannakopoulos, "pyaudioanalysis: An open-source python library for audio signal analysis," *PloS one*, vol. 10, no. 12, 2015.